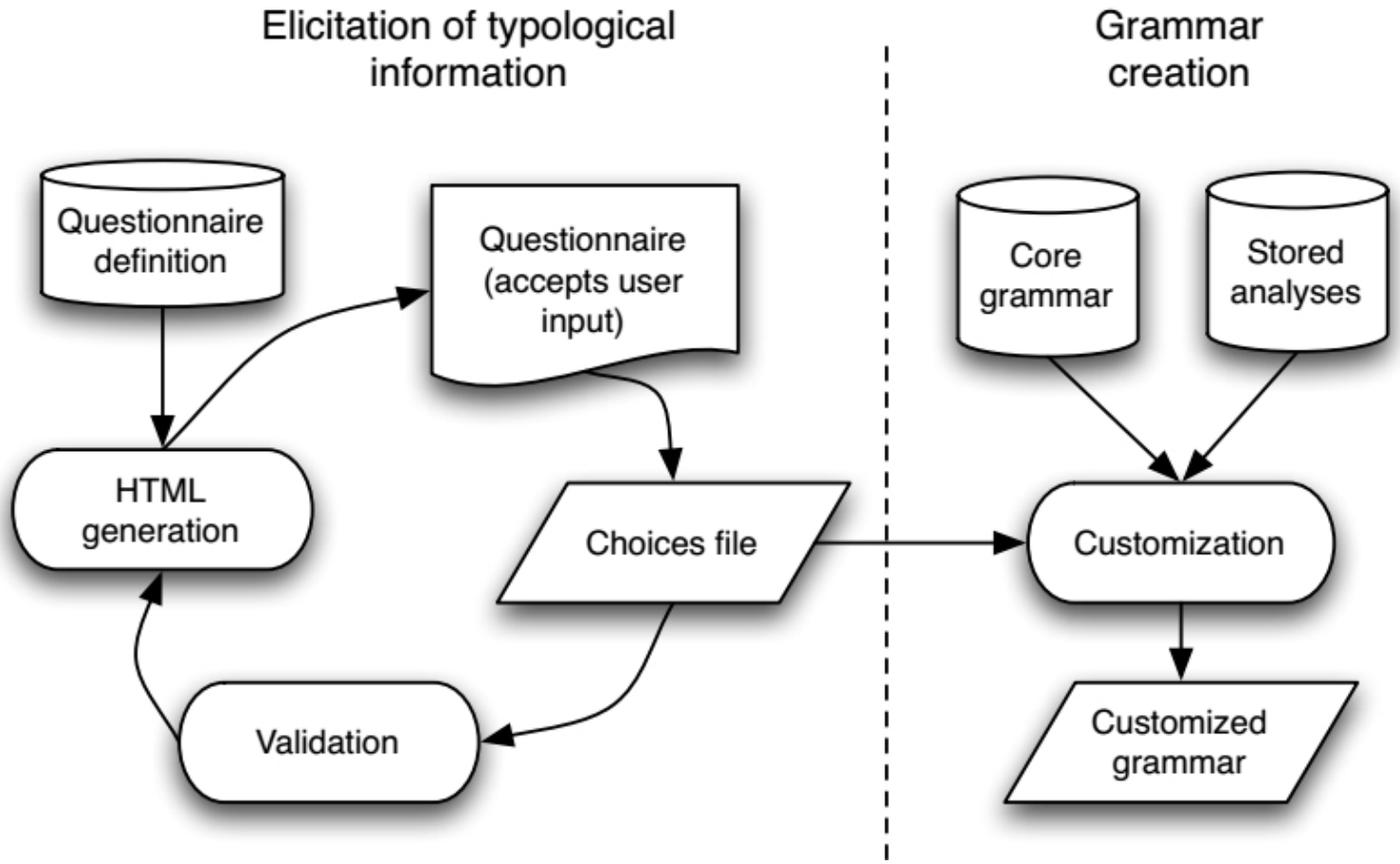
MOM
Matrix-Oriented Morphological Analyzer (update)

Olga Zamaraeva
University of Washington
2015

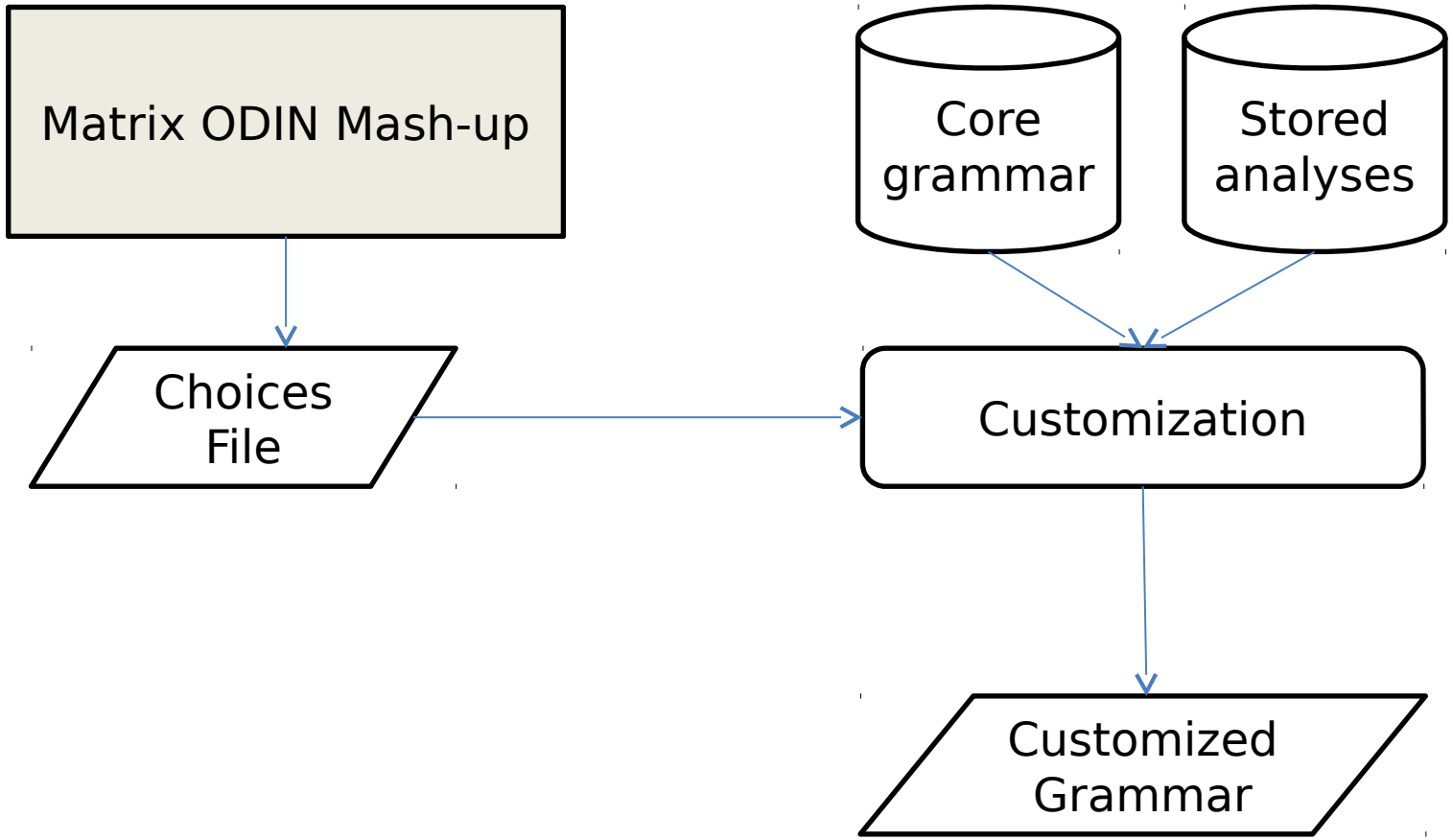
The Grammar Matrix
From webQ to TDL to LKB



This chart is from Bender et al (2010)

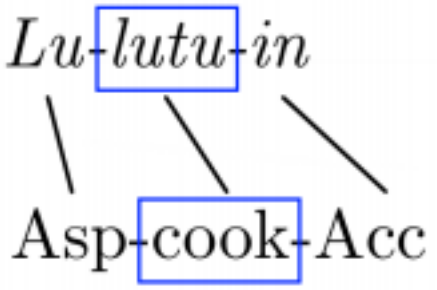
MOM: Matrix-ODIN Morphology

(From Bender, Wax, and Goodman, 2012 presentation at LSA)



MOM: originally "Matrix-ODIN Morphology"

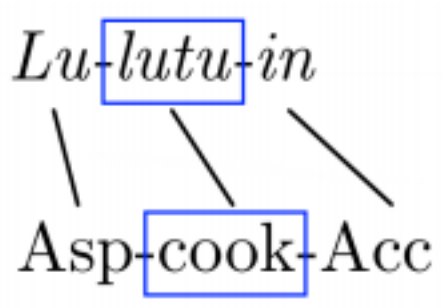
- A morphological analyzer:
  - Infer verb classes, PCs and their order (inputs)
  - Essentially a graph of verb classes and affixes




```
section=lexicon
   verb1_valence=trans
      verb1_stem1_orth=lutu
      verb1_stem1_pred=cook
```

(from Wax, 2014)

MOM: original choices file output
(from Wax, 2014)



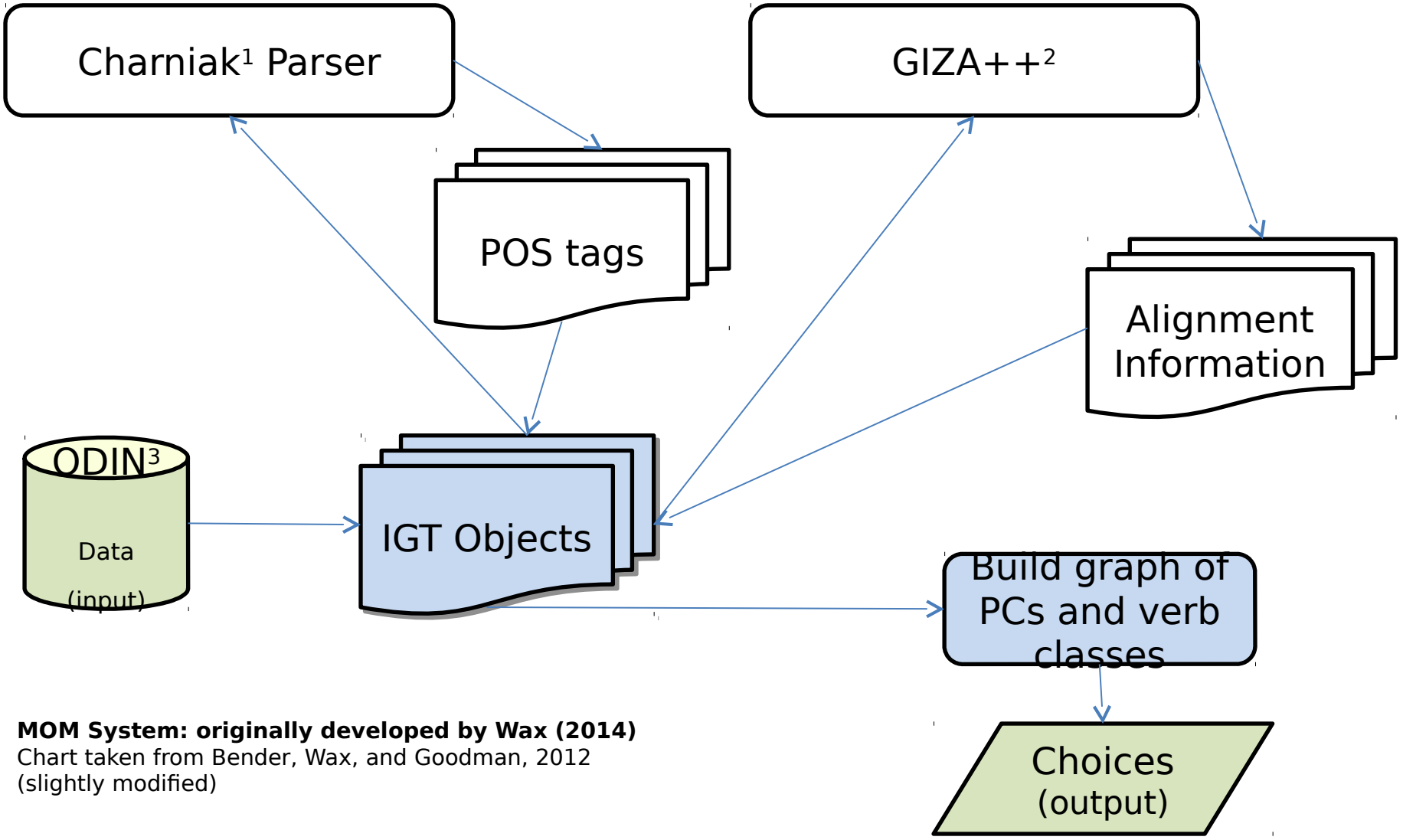

```
section=morphology
   verb-pc1_order=prefix
   verb-pc1_inputs=verb2
      verb-pc1_lrt1_feat1_name=asp
      verb-pc1_lrt1_feat1_value=asp_T
      verb-pc1_lrt1_feat1_head=verb
      verb-pc1_lrt1_lri1_inflecting=yes
      verb-pc1_lrt1_lri1_orth=lu
   verb-pc2_order=suffix
   verb-pc2_inputs=verb_pc1
      verb-pc2_lrt1_feat1_name=case
      verb-pc2_lrt1_feat1_value=acc
      verb-pc2_lrt1_feat1_head=obj
      verb-pc2_lrt1_lri1_inflecting=yes
      verb-pc2_lrt1_lri1_orth=in
```

MOM: new output (data from Language CoLLAGE)

Ivan kupi-l
sobak-u
Ivan buy-3masc.sg.PAST dog-ACC

[rus]

```
section=lexicon
 verb1_valence=trans
     verb1_stem1_orth=kupi
     verb1_stem1_pred=_buy_v_rel
 verb-pc31_inputs=verb1
   verb-pc31_lrt1_feat1_name=aspect
   verb-pc31_lrt1_feat1_value=imperfective
   verb-pc31_lrt1_feat1_head=verb
   verb-pc31_lrt1_feat2_name=tense
   verb-pc31_lrt1_feat2_value=past
   verb-pc31_lrt1_feat2_head=verb
   verb-pc31_lrt1_feat3_name=person
   verb-pc31_lrt1_feat3_value=3
   verb-pc31_lrt1_feat3_head=subj
   verb-pc31_lrt1_feat4_name=number
   verb-pc31_lrt1_feat4_value=singular
   verb-pc31_lrt1_feat4_head=subj
   verb-pc31_lrt1_feat5_name=gender
   verb-pc31_lrt1_feat5_value=masculine
   verb-pc31_lrt1_feat5_head=subj
   verb-pc31_lrt1_lri1_inflecting=yes
   verb-pc31_lrt1_lri1_orth=-l
```



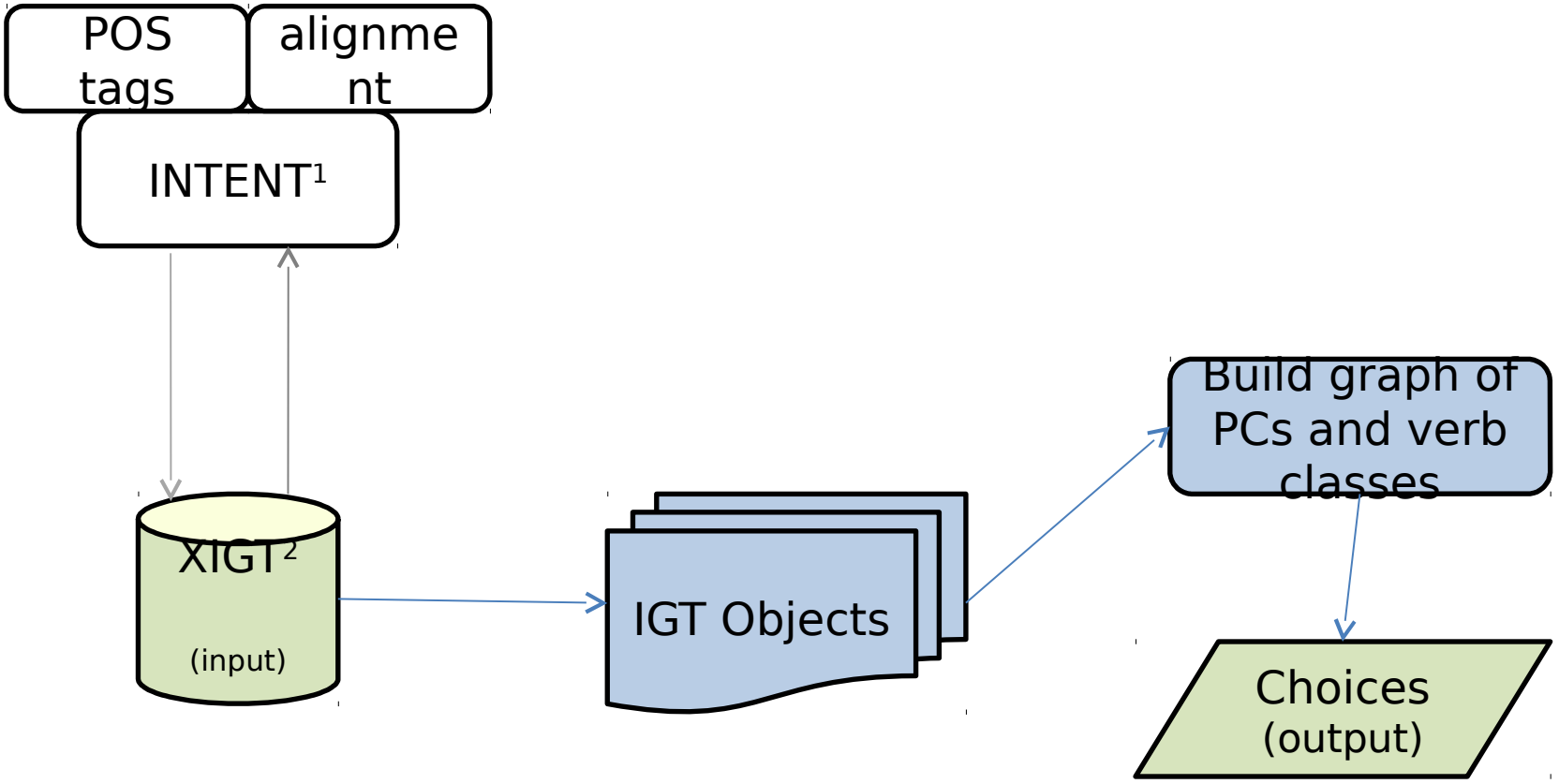
MOM System: originally developed by Wax (2014)
Chart taken from Bender, Wax, and Goodman, 2012 (slightly modified)

1. (Charniak, 1997)
2. (Och &Ney, 2000)
3. (Lewis, 2006)

Issues with original MOM

- Huge code base, very hard to debug, dependent on the system setup
- Slow
- Originally only for verbs (hard-coded)
- Ignored whatever annotation is in the data beyond ODIN format
  - And we have access to very rich corpora in ctn and mcb
- Feature Analysis incomplete

MOM System: revised



1. (Georgi, forth)
2. (Goodman et al, 2015)

Modifications

- Reduce to the core (PC graph), make it independent of other tools.
- Make XIGT-compatible
  - In order to be able to store detailed annotations
- Enrich XIGT with POS tags and alignment info separately, using INTENT, or use existing info
  - In case I have better tags than Charniak would give!

Modifications (contd.)

- Run with any POS tag combination
  - User-defined
- Make it faster (was slow due to some python bugs and duplicating some Matrix work)
  - Joshua Crowgey made some improvements
  - Don’t use .keys() unless you have to ⏪

Modifications (contd.)

- Include meaningful features in the choices
  - E.g. interpret ***3mS*** as 3rd person masculine gender constraints on the subject of the verb
  - Other interpretable features include Number, Case, Tense, Aspect… Several students worked on some of those in Emily's seminar last spring.
- These modifications were made possible by earlier modifications ⏪.

XIGT (Goodman et al., 2015)

- a data model for representing annotations of natural language data and an XML serialization of that model
- Flexible and allows very detailed annotation
- https://github.com/goodmami/xigt/wiki

INTENT (Georgi, forth.)

- The Interlinear Text Enrichment Kit
- Work in progress, but in theory does all the work needed by MOM in case the corpus does not have POS tags and alignments
- https://bitbucket.org/ryageo/intent/

Outcomes

- Emily's seminar in Spring 2015
  - Several students used new MOM to develop inference procedures for, e.g., PNG agreement
  - This is to be added to the AGGREGATION pipeline to produce richer grammars
- Matsigenka grammars
  - Discussion with Lev Michael:
    - What are the position classes?
    - Identifying mistakes in the corpus

Conclusion

- Creating grammars is a long term goal of AGGREGATION. Creating intermediate representations that are useful for analyzing data can be an important result.

- We are able to do that while ***reusing*** a lot of previous and using some ongoing (parallel) work